

USE banking_case;

select * from customer;

--First:Risk & Credit Analytics

--(1)Risk Weighting to see where the bank's money is concentrated (تصنيف حجم القروض واجمالي المديونات حسب درجه المخاطرة عشان نعرف فين بتتركز اموال البنك بالضبط (بناء علي مؤشر المخاطر

SELECT

Risk_Weighting,

COUNT(Client_ID) AS Total_Customers,

ROUND(SUM(Bank_Loans), 2) AS Total_Bank_Loans,

ROUND(SUM(Credit_Card_Balance), 2) AS Total_CC_Balance,

ROUND(AVG(Credit_Card_Balance), 2) AS Avg_CC_Balance

FROM customer

GROUP BY Risk_Weighting

ORDER BY Risk_Weighting DESC;

Client_ID	Name	Age	Location_ID	Joined_Bank	Banking_Contact	Nationality	Occupation	Fee_Structure	Loyalty_Classification	Estimated_Income	Superannuation_Savings	Amount_of_Credit_Cards	Credit_Card_Balance	Bank_Loans	Bank_Deposits	Checkin
IND01288	Raymond Mills	24	34324	2019-05-06	Anthony Torres	American	Safety Technician IV	High	Jade	75384.7734375	17677.94521875	1	484.540008544322	776242.9375	1485828.625	603617
IND65833	Julia Spencer	23	42205	2001-12-10	Jonathan Hawkins	African	Software Consultant	High	Jade	288834.3125	17398.919521875	1	2256.8798828125	1270615.375	641482.8125	229521
IND47499	Stephen Murray	27	7314	2010-01-25	Anthony Berry	European	Help Desk Operator	High	Gold	163935.234375	42825.8984375	2	4568.740234375	1052715.875	1033401.5625	652674
IND72498	Virginia Garza	40	34594	2019-03-28	Steve Diaz	American	Geologist II	Mid	Silver	356808.125	5473.14990234375	2	4205	121195.0625	1048157.5	104815
IND60181	Melissa Sanders	46	41269	2012-07-20	Shawn Long	American	Assistant Professor	Mid	Platinum	130711.6796875	48077.6015625	1	3779.48899023438	1048301.9375	487782.53125	446644
IND78532	Samuel Hudson	23	13204	2019-02-07	Douglas Tucker	American	Help Desk Technician	High	Silver	118326.9609375	6028.2001953125	2	5148.56005859375	601902.5	1307269.375	745627
IND95683	Timothy Alexander	46	42910	2002-06-02	Douglas Tucker	Asian	Account Coordinator	High	Gold	57336.46875	13829.16015625	1	959.900024414063	208909.6875	41200.1798875	605881
IND40785	Carl Martin	78	6127	2000-11-03	Steve Diaz	European	Automation Specialist II	Mid	Gold	65125.80078125	31077.80078125	1	4575.580078125	1140704.75	156983.125	53889

Risk_Weighting	Total_Customers	Total_Bank_Loans	Total_CC_Balance	Avg_CC_Balance
5	160	156034658.72	847977.93	5299.86
4	322	273278143.7	1450325.54	4504.12
3	460	361852057.66	1890826.06	4110.49
2	1222	703608971.71	3785786.96	3081.66
1	82	77090459.76	1577761.54	1924.13

--(2)Top High-Risk Debtors

--The most dangerous and top indebted

customers(لاستخراج قائمة بالعملاء ذوي المخاطر العالية (مثلاً customers
(.الدرجة 4 و 5) والذين يمتلكون قروضاً ضخمة

SELECT TOP 50

**Client_ID, Name, Occupation, Risk_Weighting,
Bank_Loans, Credit_Card_Balance**

FROM customer

WHERE Risk_Weighting IN (4, 5)

ORDER BY Bank_Loans DESC;

	Client_ID	Name	Occupation	Risk_Weighting	Bank_Loans	Credit_Card_Balance
1	IND23172	Ryan Mitchell	Programmer Analyst I	5	2667556.75	7439.35986328125
2	IND22626	Terry Chavez	Geologist IV	4	2584311.25	8833.650390625
3	IND99978	Russell Wa...	Operator	5	2538730.25	13749.830078125
4	IND99607	Sean Pierce	Health Coach II	5	2439164.25	1871.05004882813
5	IND70050	Ralph Sulliv...	Statistician II	4	2381138.25	8095.64990234375
6	IND19422	Sean Harris...	Programmer III	4	2374318.5	9810.3798828125
7	IND70565	Craig Hansen	Research Assistant II	5	2349148.25	1693.08996582031
8	IND31640	Larry Cooper	VP Marketing	5	2348592	7977.9501953125

--(3)Business vs Personal Loans Analysis of the proportion of business debt to personal loans

--To know is the bank more focused on corporate and business funding_lending or individuals(تحليل نسبة الديون التجارية إلى القروض الشخصية لمعرفة هل البنك يركز أكثر على تمويل الشركات والأعمال أم الأفراد)

SELECT

Occupation,

ROUND(SUM(Business_Lending), 2) AS

Total_Business_Lending,

ROUND(SUM(Bank_Loans), 2) AS Total_Personal_Loans

FROM customer

GROUP BY Occupation

ORDER BY Total_Business_Lending DESC;

	Occupation	Total_Business_Lending	Total_Personal_Loans
129	Staff Accountant IV	11348554.99	7593231.91
130	Programmer II	11316799.13	4722063.51
131	Staff Scientist	11299714.16	10569779.82
132	Food Chemist	11217069.91	7671318.37
133	GIS Technical Architect	11125309.22	7707078.58
134	Research Nurse	11083192.59	9407835.63
135	Senior Cost Accountant	10936272.31	6323329.36
136	Budget/Accounting Analy...	10858126.45	10099999.02

--(4)checking Accounts to Customers who have indebted their credit cards revealed their current accounts A strong risk index (client is distracting from the Credit Card more than he has in his current account checking_accounts)

كشف حسابات العملاء التي بيصرفوا اموال من الكريديت كارد اكثر من التي (بيملكوه في حساباتهم الجاريه)

SELECT

**Client_ID, Name, Checking_Accounts,
Credit_Card_Balance,**

**ROUND((Credit_Card_Balance - Checking_Accounts), 2)
AS Deficit_Amount**

FROM customer

WHERE Credit_Card_Balance > Checking_Accounts

ORDER BY Deficit_Amount DESC;

	Client_ID	Name	Checking_Accounts	Credit_Card_Balance	Deficit_Amount
1	IND72949	Robert Harrison	0	11910.990234375	11910.99
2	IND57917	Jane Hicks	0	7274.740234375	7274.74
3	IND41393	Barbara Smith	0	6875.06005859375	6875.06
4	IND13072	Phillip Banks	0	6121.10009765625	6121.1
5	IND20489	Anthony Parker	0	6034.64013671875	6034.64
6	IND66550	Patricia George	0	5841.580078125	5841.58
7	IND96395	Joshua Pierce	0	4593.75	4593.75
8	IND55634	Scott Jones	0	4447.330078125	4447.33

--Second:Wealth Management & Deposits

--(1)Total cash liquidity per customer (Total Liquidity)

حساب إجمالي المبالغ السائلة للعميل في الحسابات الجارية، التوفير، والعملات (الأجنبية)

SELECT TOP 100

Client_ID, Name,

ROUND((Checking_Accounts + Saving_Accounts + Foreign_Currency_Account), 2) AS Total_Liquid_Cash,

Loyalty_Classification

FROM customer

ORDER BY Total_Liquid_Cash DESC;

	Client_ID	Name	Total_Liquid_Cash	Loyalty_Classification
2	Click to select all grid cells		2062926.38	Silver
30	IND47218	Samuel Hicks	2057475.63	Silver
31	IND19935	William Perry	2056521.59	Platinum
32	IND92553	Daniel Tho...	2053598.85	Jade
33	IND82767	Daniel Gom...	2037230.52	Jade
34	IND88473	Margaret W...	2029920.53	Jade
35	IND32547	Wayne Hart	2021677.53	Jade
36	IND70365	Billy Wagner	2009761.72	Gold

--(2)Foreign_Currency_Account

--Analysis of customer preference for foreign versus domestic currencies (لمعرفة أي الفئات أو الجنسيات تفضل الاحتفاظ بأموالها في حسابات العملات الأجنبية)

SELECT

Nationality,

ROUND(SUM(Foreign_Currency_Account), 2) AS

Total_Foreign_Currency,

ROUND(SUM(Saving_Accounts + Checking_Accounts),

2) AS Total_Local_Currency

FROM customer

GROUP BY Nationality

ORDER BY Total_Foreign_Currency DESC;

	Nationality	Total_Foreign_Currency	Total_Local_Currency
1	European	39491782.68	725250553.73
2	Asian	22936814.66	423027230.06
3	American	14592457.15	280446582.94
4	Australian	7036054.15	137593598.25
5	African	5593481.36	95685941.71

--(3)Bank_Deposits By Properties_Owned

--Rating Wealths based on the number of owned properties(مقارنة مدخرات العملاء بناءً على عدد الأملاك والعقارات التي يمتلكونها)

SELECT

Properties_Owned,

COUNT(Client_ID) AS Customer_Count,

ROUND(AVG(Bank_Deposits), 2) AS Avg_Bank_Deposits,

ROUND(AVG(Estimated_Income), 2) AS Avg_Income

FROM customer

GROUP BY Properties_Owned

ORDER BY Properties_Owned DESC;

	Properties_Owned	Customer_Count	Avg_Bank_Deposits	Avg_Income
1	3	742	644907.42	171686.6
2	2	777	698634.22	169682.34
3	1	776	656519.97	169562.47
4	0	705	686327.65	174609.91

--8: VIP customer chip (High-Deposit owners and low-debits) (استخراج قائمة العملاء المثاليين للبنك التي يملكون ودائع عالية جداً) (وبدون مخاطر

SELECT

**Client_ID, Name, Bank_Deposits, Bank_Loans,
Risk_Weighting**

FROM customer

**WHERE Bank_Deposits > 500000 AND Bank_Loans = 0
AND Risk_Weighting <= 2**

ORDER BY Bank_Deposits DESC;

	Client_ID	Name	Bank_Deposits	Bank_Loans	Risk_Weighting
1	IND90914	Scott Phillips	2542970.75	0	2
2	IND73620	Robert Stewart	1927415.875	0	2
3	IND32633	Gerald Lee	1517817.5	0	2
4	IND76462	James Collins	1333804.125	0	2
5	IND79637	Eugene Boyd	1268649.125	0	2
6	IND63102	Edward Tuc...	724809.5	0	1
7	IND80929	Mary Austin	687595.125	0	2
8	IND72516	Beverly Stew...	561513.3125	0	2
9	IND33193	Scott Ward	502316.84375	0	2

--Third:Segmentation & Marketing Analytics

--(1) Gender-Based Financial Behavior (تحليل متوسط الدخل)
(والودائع والمديونية بناءً على نوع الجنس لتخصيص الحملات التسويقية)

SELECT

GenderId,

COUNT(Client_ID) AS Total_Count,

ROUND(AVG(Estimated_Income), 2) AS Avg_Income,

ROUND(AVG(Bank_Deposits), 2) AS Avg_Deposits,

ROUND(AVG(Credit_Card_Balance), 2) AS

Avg_CC_Balance

FROM customer

GROUP BY GenderId;

	GenderId	Total_Count	Avg_Income	Avg_Deposits	Avg_CC_Balance
1	1	1488	175379.07	669357.17	3179.47
2	2	1512	167295.66	673728.25	3172.99

معرفة أي الفروع أو المناطق تمتلك أعلى (2) Branch/Location ID Analysis
(حجم إيداعات وأعلى حجم قروض)

SELECT

Location_ID,

COUNT(Client_ID) AS Total_Customers,

ROUND(SUM(Bank_Deposits), 2) AS Total_Deposits,

ROUND(SUM(Bank_Loans), 2) AS Total_Loans

FROM customer

GROUP BY Location_ID

ORDER BY Total_Deposits DESC;

	Location_ID	Total_Customers	Total_Deposits	Total_Loans
129	25946	1	2178976.75	1103688.88
130	25969	1	2178799	678417.69
131	37953	1	2156888.5	1035646.75
132	25853	1	2150327.75	442208.53
133	11581	1	2146327.25	1495246
134	41327	1	2140826.25	1275318.63
135	38823	1	2140190.25	782563
136	15809	1	2137439.25	122158.61
137	19265	2	2137064.13	2053648.81
138	13708	1	2134158.5	1763699.13
139	24291	1	2129776.75	479875.31
140	6166	2	2128230.88	1330141.31
141	1987	2	2118348.72	2501577.81
142	25267	1	2114375.75	1141430.63
143	15696	1	2113412.25	946099.75
144	23363	1	2113374.25	1158128.5
145	37276	1	2112808.75	1577746.25
146	19145	1	2109610	251052.59
147	38792	1	2101187	1484824.13
148	17836	1	2100661	356455.56

--(3)Cohort Analysis

--Analysis of loyalty and customer retention rate based on the

Join Date(تحليل الولاء ومعدل الاحتفاظ بالعملاء بناءً على تاريخ الانضمام ومعرفة
(متوسط فترة بقاء العملاء وحجم أصولهم بناءً على سنة انضمامهم للبنك)

SELECT

YEAR(Joined_Bank) AS Join_Year,

COUNT(Client_ID) AS New_Customers,

ROUND(AVG(Bank_Deposits), 2) AS Avg_Deposits

FROM customer

GROUP BY YEAR(Joined_Bank)

ORDER BY Join_Year ASC;

	Join_Year	New_Customers	Avg_Deposits
1	1995	97	600296.63
2	1996	109	817951.09
3	1999	101	728673.12
4	2000	99	591491.53
5	2001	106	672082.22
6	2002	110	593674.39
7	2003	98	707068.49
8	2004	106	664232.18
9	2005	119	635206.56
10	2006	112	678835.19
11	2007	110	684465.04
12	2008	117	748774.63
13	2009	105	645263.31
14	2010	99	724420.93
15	2011	90	719836.53
16	2012	109	673823.71
17	2013	126	610303.3
18	2014	119	689667.45

--Fourth: (1)KPIs & Financial Ratios(تحليلات الأداء والنسب المالية)

--(Loan-to-Deposit Ratio - LDR) (نسبة القروض إلى الودائع لكل عميل لقياس)
(مدي اعتما العميل علي السلف مقارنة بمدخراته ف الحساب)

SELECT

Client_ID, Name,

ROUND((Bank_Loans / NULLIF(Bank_Deposits, 0)) * 100, 2) AS
Loan_to_Deposit_Percentage

FROM customer

WHERE Bank_Deposits > 0

ORDER BY Loan_to_Deposit_Percentage DESC;

	Client_ID	Name	Loan_to_Deposit_Percentage
1	IND46526	Patrick Sims	2778.23
2	IND68994	Marilyn Coleman	2576.75
3	IND95530	Raymond 'War...	2569.08
4	IND83050	Kenneth 'Walker	2521.3
5	IND70183	Jesse Lane	2488.36
6	IND14819	Mark Ward	2407.7
7	IND21547	Mark Owens	2329.68
8	IND40038	Kimberly Young	2318.85
9	IND18126	Susan Thomp...	2224.47
10	IND60476	Carolyn Ross	2180.72
11	IND76682	Nicole Hayes	2175.28
12	IND49097	Joyce Gilbert	2073.35
13	IND69570	Stephanie Th...	2065.86
14	IND36864	Rebecca Han...	2009.28
15	IND42511	Annie 'Washin...	1981.85
16	IND70754	Justin Ramirez	1947.91

--(2)Total Asset Under Management - AUM (حساب إجمالي أصول البنك لدى العملاء لمعرفة مجموع كل الحسابات الإيجابية (التي تمثل أموالاً داخل البنك

SELECT

ROUND(SUM(Checking_Accounts), 2) AS

Total_Checking,

ROUND(SUM(Saving_Accounts), 2) AS

Total_Savings,

ROUND(SUM(Foreign_Currency_Account), 2) AS

Total_Foreign,

**ROUND(SUM(Checking_Accounts +
Saving_Accounts + Foreign_Currency_Account), 2)
AS Overall_Bank_AUM**

FROM customer;

	Total_Checking	Total_Savings	Total_Foreign	Overall_Bank_AUM
1	963278846.12	698725060.56	89650590	1751654496.67

--(3)Real Estate Investment Ratio compared to Annual Income(تحليل أصحاب العقارات لمعرفة هل زيادة عدد العقارات المملوكة (بتؤثر علي الدخل المقدر

SELECT

Properties_Owned,

COUNT(Client_ID) AS Total_Clients,

**ROUND(AVG(Estimated_Income), 2) AS
Avg_Annual_Income**

FROM customer

GROUP BY Properties_Owned;

	Properties_Owned	Total_Clients	Avg_Annual_Income
1	0	705	174609.91
2	3	742	171686.6
3	1	776	169562.47
4	2	777	169682.34

--(Fifth) Arrange the Branchs (ترتيب الفروع حسب إجمالي القروض)
التجارية مع حساب النسبة المئوية المساهمة في كل فرع مع توزيع حجم إقراض
(الأعمال علي مستوى الفروع ونسبة كل فرع من الإجمالي

SELECT

**Location_ID, ROUND(SUM(Business_Lending), 2) AS
Branch_Business_Lending,**

**ROUND((SUM(Business_Lending) /
SUM(SUM(Business_Lending)) OVER()) * 100, 2) AS
Contribution_Percentage**

FROM customer

GROUP BY Location_ID

ORDER BY Branch_Business_Lending DESC;

	Location_ID	Branch_Business_Lending	Contribution_Percentage
65	23240	2700944	0.1
66	36151	2698297.75	0.1
67	11917	2695486.06	0.1
68	37875	2680075.25	0.1
69	11596	2655115	0.1
70	14000	2650090.85	0.1
71	11356	2635585.5	0.1
72	31801	2633258.5	0.1

--(2)Classified customers into (percentiles) segments based on savings accounts balances

تقسيم العملاء إلى 4 مجموعات متساوية بناء على أرصده حسابات التوفير ويكون (--
(التقسيم كالتالي (الـ 25% الأغنى، ثم الأقل فالأقل)

SELECT

Client_ID, Name, Saving_Accounts,

NTILE(4) OVER(ORDER BY Saving_Accounts DESC) AS
Savings_Quadrant

FROM customer;

	Client_ID	Name	Saving_Accounts	Savings_Quadrant
1	IND70565	Craig Hansen	1724118.375	1
2	IND90712	Michael Patterson	1712270.25	1
3	IND95974	Joseph Grant	1661769.875	1
4	IND92500	Daniel Gomez	1574475.75	1
5	IND86149	Martin Marshall	1566778.125	1
6	IND97484	Ralph Campbell	1549728.875	1
7	IND53236	John Frazier	1507703.625	1
8	IND75973	James Grant	1452705.5	1
9	IND96588	Daniel Johnson	1426470.75	1
10	IND99554	Russell Coleman	1422244.375	1
11	IND83480	Bobby Reed	1414641.875	1
12	IND81510	Harry Burns	1385917.625	1
13	IND79704	Harold Marshall	1371927	1
14	IND92553	Daniel Thompson	1368697.5	1
15	IND98787	Roy George	1349469.25	1

--(3)Outliers Detection (استخراج العملاء الذين يمتلكون أرصدة شاذة أو ضخمة)
 جداً وتتجاوز حساباتهم الجارية متوسط الحسابات الجارية في البنك بـ 3 أضعاف الانحراف
 (المعياري)

WITH Stats AS (

SELECT AVG(Checking_Accounts) AS Avg_Check,
 STDEV(Checking_Accounts) AS StDev_Check FROM customer
)

SELECT Client_ID, Name, Checking_Accounts

FROM customer, Stats

WHERE Checking_Accounts > (Avg_Check + (3 * StDev_Check))

ORDER BY Checking_Accounts DESC;

	Client_ID	Name	Checking_Accounts
1	IND68515	Ronald Reed	1969923.125
2	IND47511	Johnny Patterson	1867963.5
3	IND99522	Louis Hayes	1829497.375
4	IND19422	Sean Harrison	1800437.5
5	IND19935	William Perry	1768586.125
6	IND76909	Frank Fox	1716707.625
7	IND57702	Fred Cole	1702050.75
8	IND91720	Harry Gibson	1676896.375
9	IND85031	Philip Nelson	1654140.375
10	IND56121	Daniel Crawford	1599365.875

--(4)Executive Summary KPI

SELECT

COUNT(Client_ID) AS Total_Active_Clients,

**ROUND(SUM(Checking_Accounts + Saving_Accounts),
2) AS Total_Local_Deposits,**

**ROUND(SUM(Bank_Loans + Business_Lending), 2) AS
Total_Active_Lending,**

**ROUND(AVG(Risk_Weighting), 2) AS
Portfolio_Average_Risk**

FROM customer;

	Total_Active_Clients	Total_Local_Deposits	Total_Active_Lending	Portfolio_Average_Risk
1	3000	1662003906.68	4374437895.1	2